MIL-C-83443A
30 May 1975

SUPERSEDING
MIL-C-83443(USAF)
22 April 1974

MILITARY SPECIFICATION

CELLS, SOLAR, SILICON, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for silicon solar cells with transparent covers (see 6.1).

1.2 Classification. Solar cells supplied to this specification shall be of the type detailed in the applicable specification sheet.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-3464	Desiccants, Activated, Bagged Packing Use and Static Dehumidification
MIL-S-19500	Semiconductor Device, General Specification For
MIL-F-22191	Barrier Materials, Transparent, Flexible, Heat Sealable
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical Electronic Instrument Communication, and Radar for Aerospace and Control Systems, Procedures For

STANDARDS

Federal

FED-STD-147	Tape, Packaging, Adhesive & Gummed, Methods of Inspection, Sampling and Testing

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 6116

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

E-490 Specification for the Solar Constant and Air Mass Zero Spectral Irradiance

(Application for copies should be sent to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet.

3.2 Qualification. The solar cells furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids. (see 6.3)

3.3 Materials

3.3.1 Cell material. The solar cells shall be produced from single crystal silicon as described in the applicable specification sheet.

3.3.1.1 Base resistivity. The base resistivity of the silicon shall be as specified in the applicable specification sheet.

3.3.2 Electrical contact and gridline material. The material for electrical contacts and gridlines shall be as specified in the applicable specification sheet.

3.3.3 Radioactive materials. No radioactive materials shall be used as parts of the solar cell, nor in the manufacture or processing of the solar cells.

3.3.4 Fluorescent or phosphorescent materials. All solar cells shall be free of materials that emit visible light of any wavelength during or immediately after illumination with ultraviolet or infrared radiation.

3.3.5 Chemical and physical properties. No materials shall be used in the manufacture of cells which may inhibit curing of the specified adhesives. Coatings on the front and rear cell surfaces shall be capable of adhering to cell after assembly as specified herein and exposure to the environments of 3.5.2, 3.5.3 and 3.5.5.

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3.3.6 Radiometric properties. The average radiometric properties of the cells shall be within the limits specified in the applicable specification sheet.

3.3.7 Material compatibility. All materials used in fabricating the component shall have compatible thermal expansion characteristics or provisions included to compensate for differences.

3.3.8 Moisture and fungus resistance. Materials which may be exposed to climatic extremes for extended periods shall resist fungus and have negligible moisture absorption. Protective coating shall not be acceptable as moisture and fungus preventatives on parts that will lose the coating during inspection maintenance and periodic inspection. Materials that are not nutrients for fungi shall be used.

3.4 Design and construction

3.4.1 Uniformity of product. All solar cells shall be fabricated identically in procedure and material to the units which pass the qualification as specified in 4.3. No major changes in materials, fabrication techniques or procedures, assembly, inspection, or testing shall be made for subsequent units without notifying the procuring activity of the change and the effectivity of the change. In addition, the solar cells shall be manufactured and processed with an approved Process Identification Document (PID) which must comply with 4.3.2.1 of this specification. A major change is defined as one requiring a revision of the PID.

3.4.2 Durability of anti-reflective coating. Anti-reflective coating may be applied to the front side of the silicon before the cover is applied so long as it does not deteriorate, separate from the cell, or degrade any of the cell properties as a consequence of any of the specified assembly, storage, test, or environmental exposures.

3.4.2.1 Anti-reflective coating defects. The cell anti-reflective coating shall have no voids longer than 1.80 mm in diameter and no more than 5 voids per cm^2 between 0.60 and 1.80 mm in diameter. Voids less than 0.60 mm in diameter may usually be ignored except for possible consideration under paragraph 3.10, Workmanship.

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3.4.3 Solar cell junction area. The solar cell junction area shall be located only at the upper silicon surface which carries the top collector grid. The junction area shall not extend over or wrap around any other edges of the surface The junction shall be formed in accordance with the processes described in the PID.

3.4.4 Electrical output before cover glass installation. The application of the cover glasses to completed cells in accordance with 4.5.15 shall not alter the electrical output to values lower than those specified in the applicable specification sheet.

3.4.5 Contacts and gridlines. Solar cell contacts shall be silver (Ag-Ti or Ag-Pd-Ti) or aluminum. Silver contacts shall be solderable or weldable to cell interconnects or test tabs. Aluminum contacts shall be weldable to interconnects or test tabs. Soldered interconnections are suitable for use at temperatures up to 250°F. Welded interconnections may be used for any temperatures which will not degrade cell performance below the requirements specified herein. Aluminum contacts are appropriate to meet nuclear hardening requirements.

3.4.5.1 Contact and gridline configurations. Front and back surface contacts shall be sized and located in accordance with figure 1, 2, or 3, whichever is applicable. Gridlines shall be sized and spaced so as to provide an optimum compromise between (1) the series resistance losses of the cell, and (2) the loss in active cell area resulting from covering the active area with grids.

3.4.5.1.1 Sufficient metalization paths shall be provided between interconnecting pads such that 0.1 ohm or less resistance is measured between adjacent pads.

3.4.5.2 Thickness and finish. The contacts and gridlines shall be 1.5 to 3.0 µm thick and shall have surface finished 75 nanometers RMS maximum, for welded interconnection.

3.4.5.2.1 Cells for solder interconnection shall have contact surface finish prior to solder application of 200 nanometers RMS maximum.

3.4.5.3 Contact appearance. There shall be no pinholes greater than 0.3 mm diameter penetrating the contact exposing either submetal or silicon. There shall be no evidence of contact peel or delamination when visually inspected with a microscope having 50 to 10 power magnification.

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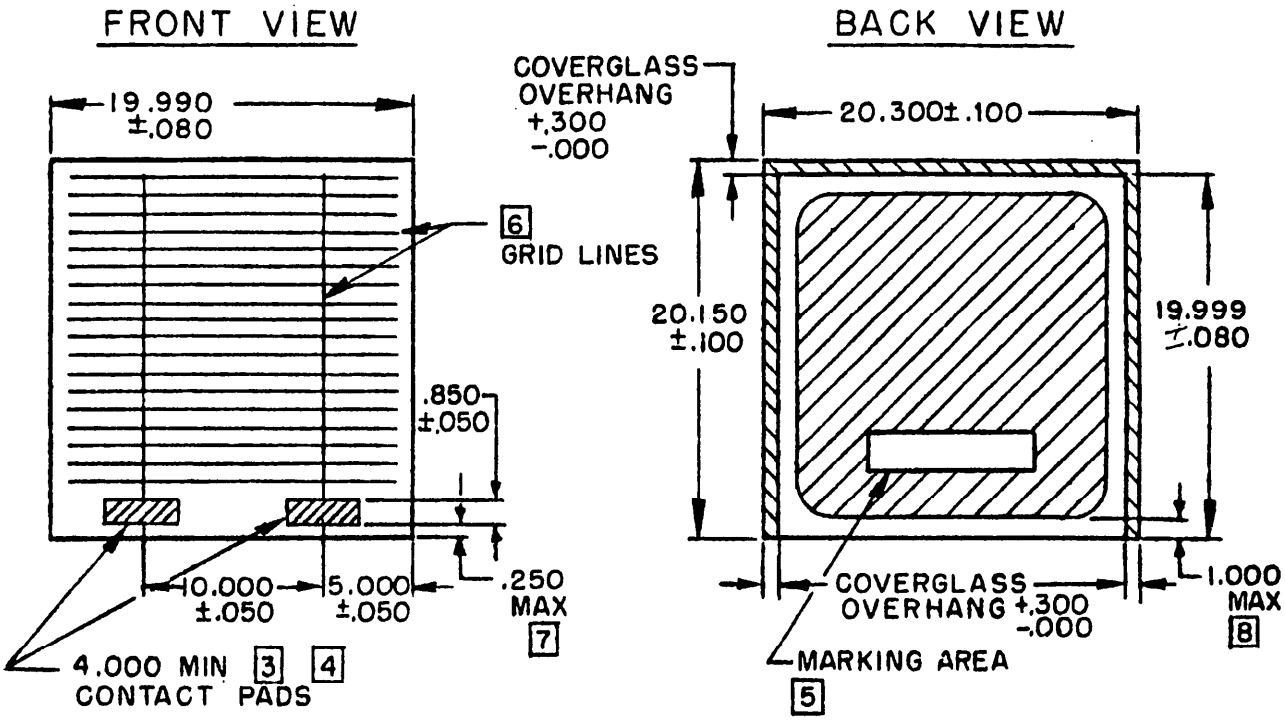


Figure 1. Dimensions 2.0 cm x 2.0 cm Silicon Solar Cell

NOTES

1. Dimensions in millimeters
2. Finished cells shall be capable of seating flat within a cavity 20.14 mm x 20.14 mm max and an angularity of 0° $\pm$20 minutes
3. Top contact pad may be full length of the cell
4. Contact pads to be free of optical coating
5. Marking as specified in 3.8 legible with naked eye
6. Gridlines should not extend closer than 0.25 mm from the cell edge. Number and shape of gridlines optional (see 3.4.5.1)
7. Maximum contact free area
8. Maximum bare silicon frame
9. Thickness 0.20 $\pm$0.05 silicon only. Contacts adhesive and cover will add to this dimension
10. Front view without coverglass. Coverglass should cover all front surface except contact pad and may extend over all cell edges except contact edge.

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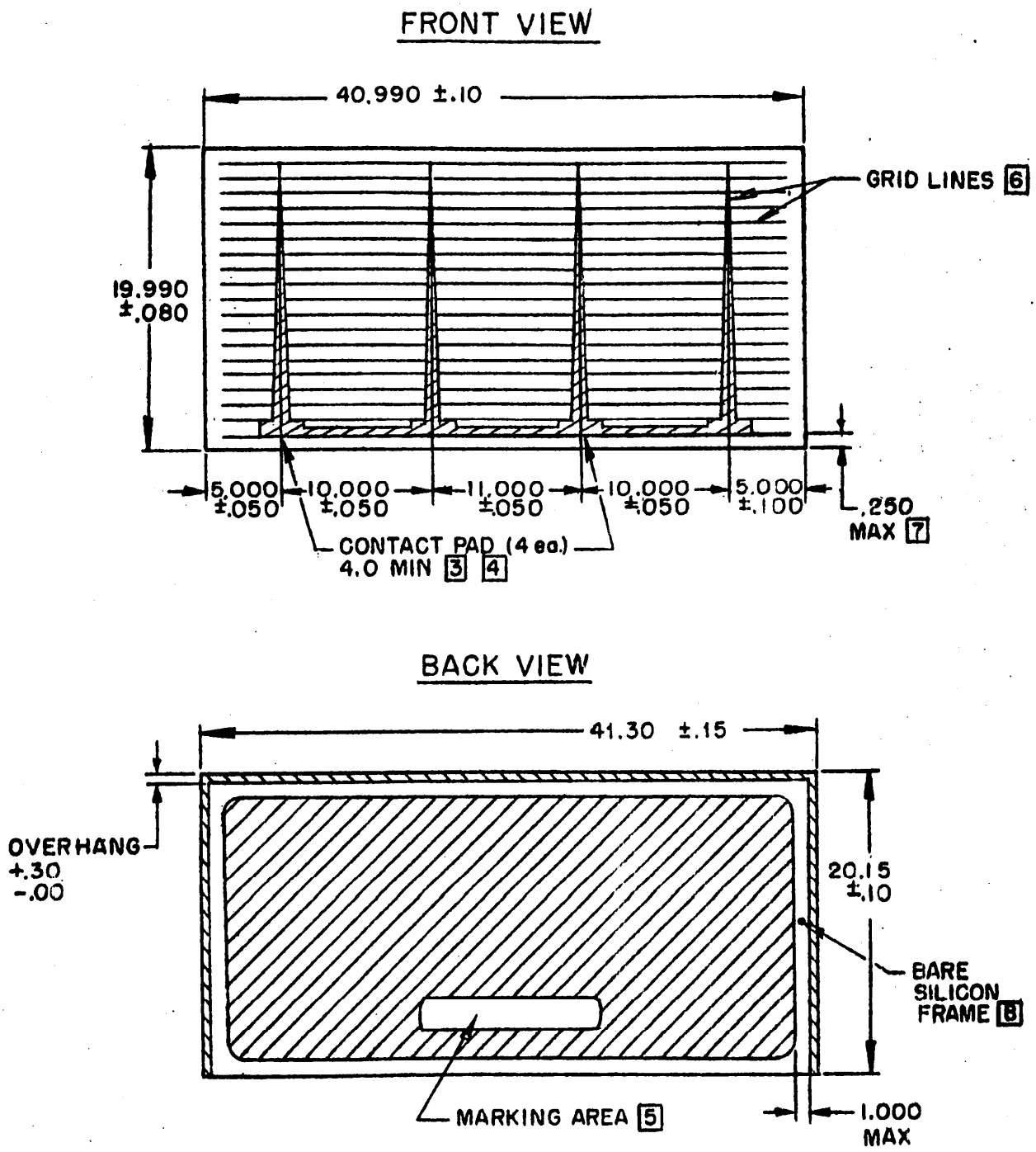


Figure 2. Dimensions of 2.0 x 4.1 cm Solar Cell

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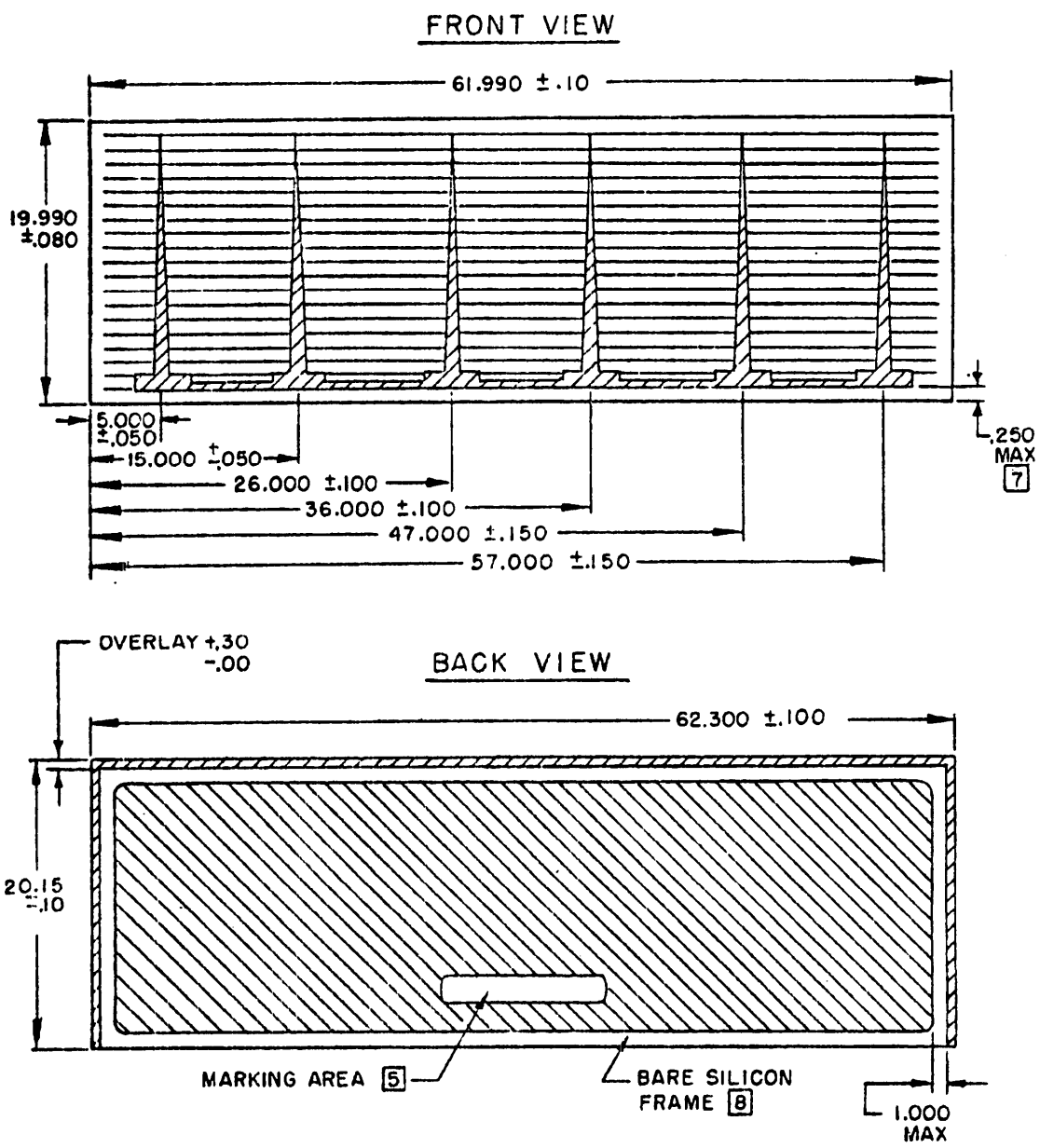


Figure 3. Dimemsions of 2.0 x 6.2 cm Cell

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3.4.5.4 Solderable contacts. The contacts and gridlines shall be coated (tinned) with a solder alloy of approximately 62 percent tin, 36 percent lead, and 2 percent silver, per MIL-S-45743. The solder coating shall be of uniform thickness not to exceed 80µm above the surface of the silicon. The solder on the back contact shall be a matted surface. The solder coating shall cover a minimum of 95 percent of all contact and of all grid area.

3.4.5.5 Contact integrity after high temperature exposure. Contacts and gridlines on cells without cover glass installed, with solder coating if applicable, shall be capable of withstanding 120°C for 15 minutes and a subsequent tape peel test as defined in 4.5.4 The high temperature exposure and tape peel shall result in < 2 percent short circuit current degradation. A cell shall have failed the peel test if more than 5 percent of the top contact or 10 percent of the bottom contact is removed by the tape. Any solder dewetting in excess of 10 percent of the contact area shall constitute failure.

3.4.5.6 Contact solderability. Each solderable contact area shall be capable of having a metallic pull tab soldered to it in accordance with the procedure in 4.5.5.4. The metal for this tab is specified in the applicable specification sheet.

3.4.5.6.1 Soldered contact mechanical integrity. Each solderable contact area after being soldered according to 3.4.5.6 shall possess a pull strength of 6 Newton (600 gm).

3.4.5.7 Contact weldability. Each weldable contact area shall be capable of having a metallic pull tab welded to it in accordance with the procedure in 4.5.5.5. The metal for this tab is specified in the applicable specification sheet.

3.4.5.7.1 Welded contact mechanical integrity. Each weldable contact area after being welded according to 3.4.5.7 shall possess a pull strength of 2.5 Newton (250 gm).

3.4.5.8 Soldered contact and welded contact electrical integrity. Each solar cell shall be capable of having four metallic pull tabs either soldered to it in accordance with 3.4.5.6 or welded to it in accordance with 3.4.5.7, whichever is applicable, without degrading the cell output current more than 1.5 percent at the cell voltage specified in the applicable specification sheet. Two of the tabs shall be on front contact areas and two tabs shall be on back contact area. The average output current degradation for any sample of cells shall not exceed 1.0 percent. This output shall be based on actual values measured before and after, and shall be corrected for variations in measurement conditions in a manner mutually agreed upon by the procuring activity and the manufacturer.

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3.4.6 Chemical and physical compatibility. All solar cells supplied to this specification shall be capable of being assembled as specified in 4.5.5.4, 4.5.5.5 and 4.5.15 herein, without any further cleaning, and thereafter meet all the requirements of 3.4.5, 3.5.3 and 3.3.4 except as allowed in 3.4.5.3, 3.5.5, 3.5.4 and 3.4.5.4.

3.5 Performance

3.5.1 Electrical output. Each solar cell shall be capable of converting natural solar irradiation into the minimum electrical power required by the applicable specification sheet. Each solar cell shall be grouped according to its electrical output characteristics called out in the applicable specification sheet and shall have an electrical output not less than the minimum specified for the output group into which it is placed. The minimum average electrical output of any shipping lot is required by the applicable specification sheet. It shall be based upon the quantity of cells in each electrical group. Actual measured output of individual cells shall not be used.

3.5.2 High temperature vacuum. Each solar cell shall meet the performance requirements of this specification after being maintained at 120°C and a pressure of 10^{-5} torr or less for 168 hours.

3.5.3 Storage temperature and humidity. Each solar cell shall be capable of meeting all requirements of this specification after storage for a continuous period of 30 days at a minumum temperature of 45°C and a relative humidity of at least 90 percent, except that the permissible current output degradation at the cell voltage as specified in the applicable specification sheet shall not exceed 1.5 percent for any cell based on actual values measured before and after the exposure and after corrections are made for variation in measurement conditions by a method mutually agreed to by the procuring activity and the cell manufacturer.

3.5.4 Contact welding and temperature cycling. The solar cell shall be capable of meeting all requirements specified below after having had 4 contact pull tabs either soldered in accordance with 4.5.5.4 or welded in accordance with 4.5.5.5 and then having been subjected to 1000 temperature cycles between -195°C and +100°C with an initial cooling rate of at least 100°C per minute and any ambient pressure.

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3.5.4.1 Electrical output after temperature cycling. After welding the tabs and temperature cycling, each cell shall meet the electrical requirements specified in the applicable specification sheet, except the maximum allowed electrical output degradation at the cell voltage shall be 2.5 percent for any single cell, and 1.5 percent average for any sample of cells. The average degradation shall be based on electrical output measurements of all cells including rejected ones. Also, this degradation shall be based on actual values measured before and after the temperature cycling and after corrections have been made for variations in measurement conditions. The method for these corrections shall be mutually agreed upon by the procuring activity and the manufacturer.

3.5.4.2 Contact pull strength after temperature cycling. The contact pull tab attached to the cells prior to temperature cycling specified in 3.5.4 shall possess a 45° pull strength of 2.5 Newtons (250 gm) minimum for welded tabs and 6 Newtons (600 gm) for soldered tabs. A cell shall be considered to have failed this test if any one contact pull tab has become detached during the pull test.

3.5.4.3 Color and finish after temperature cycling. All requirements specified in 3.7 shall be met after temperature cycling.

3.5.5 Charged particle radiation resistance. Exposure of solar cells protected by cover glass to a normally incident particle fluence as specified in the applicable specification sheet shall not degrade their electrical output below that specified in the applicable specification sheet.

3.6 Dimensions and weight

3.6.1 Dimensions. The solar cell dimensions with cover slip shall be given on the specified figure 1, 2, or 3.

3.6.2 Weight. The average weight of solar cells with .006 inch (.152 mm) cover slip in each shipping lot shall not exceed 0.10 gm/cm^2. The weight of cells with thicker coverglass shall be as specified in the applicable specification sheet.

3.7 Color and finish. The appearance of each solar cell shall conform to the requirements specified in 3.7.1 and 3.7.2. Inspection criteria shall be as defined in the PID (4.3.2.1). Contact appearance criteria and functional requirements are specified in 3.4.5.

3.7.1 Color and appearance. All cells shall be reasonably uniform in color and appearance as defined in the PID.

3.7.2 Mechanical imperfections. No mechanical imperfections including chips, nicks, pinholes, voids, and blisters shall be accepted, with the exception of those items listed below. (see 6.4)

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3.7.2.1 Chips. The quantity and size of chips specified below shall not be exceeded.

Edge chip: 0.65 mm x 3.8 mm (0.025 inch x 0.150 inch) maximum

Corner chip: 1.5 mm (0.060 inch) hypotenuse maximum

No limit on quantity

3.7.2.2 Nicks. The quantity and size of nicks specified below shall not be exceeded.

Edge nick: 0.65 mm x 3.8 mm (0.025 inch x 0.150 inch) maximum

Corner nick: 1.9 mm (0.075 inch) hypotenuse maximum

No limit on quantity

3.7.2.3 Gridline discontinuities. The maximum total linear length per cell of breaks in the gridlines shall not exceed 5 percent of total gridline length. There is no limit on the number of cells with this condition. Linear breaks of 0.25 mm (0.010 inch) or less in the gridline shall be ignored.

3.8 Identification of product. Identification of individual solar cells shall consist of (1) fabrication lot number as defined in 6.4.8, and (2) current output group according to 3.5.1. The marking shall be affixed to the rear of the solar cells in the areas indicated on figures 1, 2, and 3. This identification shall be clearly readable after cleaning with water and alcohol and shall not degrade the specified mechanical, electrical, or environmental properties of the cell. The marking shall comply with 3.3.5 and all other requirements of this specification. Rubber stamping with an ink is acceptable.

3.9 Cover glass installation. The cover glass will be installed filter side down and overlapping cell edges on 3 sides. (see figures 1, 2 and 3). The cover material and adhesive shall be as stated in the applicable specification sheet.

3.10 Workmanship. All details of workmanship and finish shall be in accordance with the best practices for high quality solar cells consistent with the requirements of this specification.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. The inspection requirements specified herein are classified as follows:

a. Qualification examinations and tests (see 4.3)

b. Quality conformance examinations and tests (see 4.4)

4.3 Qualification tests. Qualification tests shall consist of the tests specified in table I. All tests shall be conducted in the sequence indicated, and shall demonstrate compliance with the respective requirements shown in table I. Unless otherwise specified all tests shall be conducted with cover glass installed as specified in the applicable specification sheet. The significant requirements for the 5 groups are as follows:

Group A - Contact welding and temperature cycling (3.5.4)

Group B - Charged particle radiation resistance (3.5.5)

Group C - Contact integrity after welding (3.4.5.7.1)

Group D - High temperature vacuum and humidity (3.5.3)

Group E - Electrical output before cover glass installation (3.4.4) and radiometric properties (3.3.6)

4.3.1 Qualification sampling

4.3.1.1 Sample size. The number of cells to be tested shall be chosen (independent of lot size) by the manufacturer according to the lot tolerance percent defective (LTPD) method in conformance to table II, so that the qualification test LTPD level is at least one level tighter than the level required for quality conformance testing. This shall apply to each of the groups of cells to be tested according to table I, except that the sample size for any group A, B, C, etc., shall not exceed 94 cells. Table II gives the minimum test sample size to assure, with a 90 percent confidence, a lot tolerance percent defective, λ, not greater than that specified. See MIL-S-19500.

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Table I. Qualification Tests

ITEM	REQUIREMENTS		TEST METHOD	GROUP A	GROUP B	GROUP C	GROUP D	GROUP E 1/
1	3.4.2	Anti-reflective coating	4.5.6					5
2	3.5.1	Electrical output	4.5.9	6	3.6	3	3,6,10	3
3	3.4.4	Electrical output before cover glass installation	4.5.15					*6
4	3.5.5	Charged particle radiation resistance	4.5.13		*5			
5	3.5.3	Storage temperature and humidity	4.5.12				*5	
6	3.5.4	Contact welding and temperature cycling	4.5.5.4 or 4.5.5.5	*8				
			4.5.11	*9				
7	3.5.4.1	Electrical output after temperature cycling	4.5.9.1	12				
8	3.5.4.2	Contact pull strength after temperature cycling	4.5.5.6	13				
9	3.5.4.3	Color and finish after temperature cycling	4.5.7					
			4.5.7.1	10				
10	3.3.6	Radiometric properties	4.5.16					*7
11	3.4.5.3	Contact appearance	4.5.7					
			4.5.7.1	3				
12	3.4.5.6. 3.4.5.7 3.4.5.8	Contact integrity after soldering or welding	4.5.5			*5		
13	3.8	Identification of product	4.5.7	7	4	4	4	4
14	3.5.1	Dimensions	4.5.8	5				
15	3.6.2	Weight	4.5.14	1				
16	3.7.1	Color and appearance	4.5.7	2	1	1	1,7	1
17	3.7.2	Mechanical imperfections	4.5.7.1	4,11	2	2	2,8,11	2
18	3.5.2	High temperature vacuum	4.5.17				9	

*Indicates significant requirement for each group per 4.3

1/ Cover glass not to be installed until step #6

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Table II. Poisson Sampling Plan for Lot Sizes Greater than 200

Acceptance Number (A)	Maximum Lot Tolerance Percent Defective (λ)							
	50	30	20	15	10	7	5	3
	MINIMUM SAMPLE SIZES							
0	5	8	11	15	22	32	45	76
1	8	13	18	25	38	55	77	129
2	11	18	25	34	52	75	105	176
3	13	22	32	43	65	94	132	221
4	16	27	38	52	78	113	158	265
5	19	31	45	60	91	131	184	308
6	21	35	51	68	104	149	209	349
7	24	39	57	77	116	166	234	390

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4.3.1.2 Selection of qualification lot. The qualification inspection lot shall be chosen from typical production lots and shall contain cells from at least 2 fabrication lots (as defined in 6.4.8) of approximately equal proportions and shall contain cells from each current output group specified in the applicable specification sheet of approximately the same proportion as required by the average output requirements also given in the applicable specification sheet.

4.3.1.3 Qualification lot size. The qualification inspection lot shall contain at least twice the number of cells required for qualification, but shall not exceed 1,200 cells.

4.3.1.4 Selection of samples. Initial samples shall be selected at random from the qualification inspection lot. When the qualification inspection lot is composed of two or more sublots, such as current output groups, the number of samples from each sublot shall be approximately proportional to the corresponding sublot size. After a test has started, the manufacturer may add an additional quantity to the initial sample, but this may be done only once for any of the groups A through E and the added samples shall be subjected to all tests within the given group. The total samples (initial and added samples) shall determine the new acceptance number. The total defectives of the initial and second sample shall be additive and shall comply with the inspection LTPD of 7 percent. The manufacturer shall retain a sufficient number of cells from the qualification lot to provide for additional samples.

4.3.2 Qualification certification. The basis for qualification shall consist of the following:

4.3.2.1 Uniformity of product. The manufacturer shall submit a Process Identification Document (PID) describing all steps used in the manufacture and test of units to meet the requirements of this specification and the applicable specification sheet. To minimize cost, and optimize delivery and quality, these steps may utilize existing procedures that have been established as a result of inhouse programs. In this regard, variations in details and sequence that comply with the spirit and intent of the requirements noted herein may be proposed. Each PID shall include, but not necessarily be limited to the following basic elements (any changes subsequent to PID release shall be approved by the procuring activity):

a. A detailed specification showing starting material parameters (including, but not limited to purity, dislocation density, doping material and level, Hall coefficient, oxygen content), junction depth, cell composition, and other data required to fully describe the cell the manufacturer intends to supply.

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b. A detailed flow chart showing all processing steps involved in the production of the device. Each step shall be identified by the title, number and sequence of its controlling document. A copy of the controlling Quality Assurance (QA) documents for each process step shall be provided. These QA documents shall be controlled by a document number and applicable revision letter.

c. A copy of the vendor's in-house product flow document. This document delineates the detailed handling for the solar cell, including quality conformance testing plans A, B, C, and qualification testing.

d. A mechanical drawing of the solar cell with cover slip which complies with this specification and reflects the configuration the manufacturer intends to supply.

e. Visual and color inspection criteria as desired by the cell manufacturer.

f. All elements of the PID shall be in reproducible form and capable of being microfilmed without loss of resolution.

4.3.2.2 Certification of conformance. Compliance with the requirements listed below shall be certified. A copy of the test data, data reduction, and analysis upon which certification is based shall accompany the Certificate of Conformance. In the event that this certification is based upon data obtained from a source other than the manufacturer, this data shall be subject to approval by the procuring activity.

Uniformity of product (3.4.1)

Cell material (3.3.1)

Material (contact) (3.3.2)

Chemical and physical compatibility (3.3.5)

Radioactive material (3.3.3)

Fluorescent and phosphorescent materials (3.3.4)

4.3.2.3 Verification of qualification. The following requirements of section 3 shall be verified at the time and place of qualification testing.

Durability of anti-reflective coating (3.4.2)

Electrical IV curve output (3.5.1)

Electrical output before cover glass installation (3.4.4)

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Charged particle radiation resistance (3.5.5)

Storage temperature and humidity (3.5.3)

Contact welding and temperature cycling (3.5.4)

Electrical output after temperature cycling (3.5.4.1)

Contact pull strength after temperature cycling (3.5.4.2)

Color and finish after temperature cycling (3.5.4.3)

Contact appearance (3.4.5.3)

Radiometric properties (3.3.6)

Contact integrity after soldering or welding (3.4.5.6)

Identification of product (3.8)

Dimensions (3.6.1)

Weight (3.6.2)

Color and appearance (3.7.1)

Mechanical imperfections (3.7.2)

High temperature vacuum (3.5.2)

4.3.2.4 Similarity. The following requirements may be verified by test or by a memorandum stating that the design has been proven by virtue of similarity to a previously qualified unit.

Solar cell junction area (3.4.3)

Table I provides a convenient cross reference concerning the various requirements, and tests.

4.3.2.5 Destructive tests. Cells submitted to destructive tests (6.4.10) shall not be submitted as partial fulfillment of any order.

4.3.2.6 Requalification. To maintain qualified status cells shall be subjected to complete qualification tests, table I, as follows:

a. At time intervals not to exceed 6 months

b. Whenever significant physical characteristic or process change is introduced.

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4.4 Quality conformance sampling and testing

4.4.1 Quality conformance sampling. Quality conformance sampling shall consist of the following:

a. In-process sampling and testing

b. Sampling plan A and testing

c. Sampling plans B and C and testing

4.4.1.1 In-process sampling and testing

4.4.1.1.1 Group plan IP-1. Each full or partial ingot or block of silicon from which solar cells are to be manufactured shall be characterized in methods chosen by the supplier or tested as required in 4.5.3 for the following parameters:

a. Dopant material, level and original form

b. Crystal orientation

c. Resistivity.

4.4.1.1.2 Group plan IP-2. Cells shall be selected at random from each fabrication lot as defined in 6.4.8 after contacting, coating and before cover glass application. The sample size shall be determined by the manufacturer according to 4.4.1.4 and 30 percent LTPD except that sample size shall not exceed 13. The units selected shall be tested as specified in table III.

4.4.1.1.3 Group plan IP-3. One-hundred percent of all completed cells shall be subjected to the in-process tests specified in table III.

4.4.1.2 Sampling plan A tests. Sample solar cells shall be selected in accordance with 4.4.1.4 from each shipping lot which has passed the in-process tests of table III. The cells selected shall then be subjected to the tests in table IV. The cells for test #8 anti-reflective coating shall be selected prior to coverglass application and not subjected to the other tests of Plan A.

4.4.1.3 Sampling plans B and C. Sample solar cells shall be selected in accordance with 4.4.1.4 from the first and last shipping lots produced for a single buy and from the shipping lots approximately 1/4, 1/2, and 3/4 of the production span. All shipping lots shall have passed the in-process tests of table III and plan A tests included in table IV.

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Table III. In-Process Acceptance Tests and Test Sequence

ITEM		REQUIREMENT	TEST METHOD	TEST SEQUENCES FOR CELLS SAMPLES PER 4.4.1	
				IP-2	IP-3
1	3.3.1	Material characterization	4.5.3		
2	3.4.2	Durability of Anti-reflective coating	4.5.6 4.5.7.1	3	
3	3.5.1	Electrical output	4.5.9.1		4
4	3.4.5.5	Contact integrity after high temperature	4.5.4	1	
5	3.4.5.6 or 3.4.5.7	Contact integrity after soldering or welding	4.5.5	2	
6	3.8	Identification of product	4.5.7		5
7	3.6.1	Dimensions	4.5.8.1 4.5.8.2		3 *
8	3.7.1	Color and Appearance	4.5.7		1
9	3.7.2	Mechanical imperfections	4.5.7.1		2

* 4.5.8.1 can be verified in the slab prior to slicing. Sequence of measurements will be described in the PID.

4.4.1.3.1 Initial group B and C test waiver. Performance of group B and C tests shall be waived for the first shipping lot when the manufacturer has conducted the applicable qualification test immediately prior to the manufacture of the first shipping lot.

4.4.1.3.2 Disposition of group B and C lots. Shipping lots from which group B and C sample cells have been selected shall not be submitted for acceptance unitl all group B and C tests have been successfully completed.

4.4.1.4 Statistical sampling. Statistical sampling shall be conducted using LTPD Method.

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Table IV. Quality Conformance Tests and Test Sequences

ITEM	REQUIREMENT		TEST METHOD	TEST SEQUENCE FOR CELLS SAMPLED PER 4.4.1		
				Sampling 1/ Plan A	Sampling 2/ Plan B	Sampling 2/ Plan C
1	3.4.2	Anti-reflective coating	4.5.6	8		
2	3.5.1	Electrical output	4.5.9.1	4	3,6	
3	3.5.1	Electrical output	4.5.9.2			6
4	3.4.4	Electrical output before cover glass installation	4.5.15			4
5	3.5.3	Humidity and temperature	4.5.12		4	
6	3.3.6	Radiometric properties	4.5.16			7
7	3.4.5.6.1 3.4.5.7.1 3.4.5.8	Contact integrity after soldering or welding	4.5.5	7		
8	3.8	Identification of product	4.5.7	5		5
9	3.6.1	Dimensions	4.5.8	3		3
10	3.6.2	Weight	4.5.14	6		
11	3.7.1	Color and appearance	4.5.7	1	1,5	1
12	3.7.2	Mechanical imperfections	4.5.7.1	2	2	2

1/ From each shipping lot per 4.4.1.2. A separate group of cells will be tested for item 1, anti-reflective coating

2/ From selected shipping lots per 4.4.1.3

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4.4.1.4.1 Sample size. The sample size shall be determined by the manufacturer (independent of lot size) in accordance with table II and a 10 percent LTPD level, except that the sample size for any plan shall not exceed 94 cells.

4.4.1.4.2 Sample selection. Samples shall be selected at random from as many different fabrication lots as possible and from each current output group approximately the same proportion as required by the average output requirement.

4.4.1.4.3 Replacements. In the event sample solar cells are accidentally damaged prior to completion of specified test sequence, each such damaged cell may be replaced by a new solar cell chosen at random from the lot under test. Each new cell shall be subjected to all the tests in the test sequence. Accidentally damaged cells which have been replaced by new units shall not be included in determining acceptability of the lot under test.

4.4.1.5 Conditional and final acceptance. Conditional acceptance shall be made only after successful completion of groups IP-1, IP-2, IP-3, and plan A tests as specified in tables III and IV. Final acceptance shall be made only after successful completion of plan B and C tests, as specified in table IV. All tests shall be conducted in the sequence indicated, and shall demonstrate compliance with the respective requirements as shown in tables III and IV.

4.4.2 Rejection and resubmittal

4.4.2.1 Rejection. Any cell which does not meet any one or more requirements of any individual test specified shall be rejected.

4.4.2.2. Rejection of fabrication lots. When a fabrication lot fails, the lot shall be scrapped and shall not be used to formulate any subsequent lot. A fabrication lot shall have failed if any of the following tests were failed:

Contact integrity after high temperature (4.5.4)

Contact integrity after soldering or welding (4.5.5)

Durability of anti-reflective coating (4.5.6, 4.5.7.1)

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4.4.2.3 Rejection and retest. When a shipping lot fails to meet the specified LTPD, all items in the lot shall be rejected. Any lot so rejected by sampling plans B or C shall require the testing of the immediate prior and next untested lots to the condition of sampling plans B and C. The procuring activity shall be immediately notified of any rejected lot.

4.4.2.4 Disposition of rejected lot. The disposition of rejected shipping lots shall be in accordance with 4.4.2.5 and 4.4.2.6.

4.4.2.5 Resubmitted lots. Lots from which defectives have been screened or reworked and which are submitted for quality conformance inspection, shall contain only devices which were in the original lot. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected for all characteristics, using tightened inspection only for the failed characteristics. No lot shall be resubmitted for plan B or C inspection after a second rejection for the same defect. No lot shall be resubmitted for plan B or C inspection after a second rejection. At the discretion of the procuring activity, testing of characteristics which are not affected by the screening process may be omitted for resubmitted lots.

4.4.2.6 Records of rejected lots. Adequate records of all rejected lots shall be maintained including lot number, quantity of cells, test data, type of test, reason for rejection and disposition, as a minimum. Two copies of each lot rejection record shall be sent to the procuring activity when disposition is made.

4.4.2.7 Disposal of samples. Solar cells subjected to destructive in-process tests and to tests under sampling plan A, B, and C shall be submitted to the procuring activity for disposition. These cells shall not be shipped on the contract or purchase order. Destructive tests are defined in 6.4.10.

4.4.2.8 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the manufacturer shall fully advise the procuring activity of all defects likely to be found and methods for correcting them.

4.4.3 Quality conformance certification. A certificate of compliance shall accompany each shipment. The certificate shall provide evidence that all the quality conformance provisions have been met.

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4.5 Test methods

4.5.1 Test methods and test conditions. The qualification and quality conformance tests required in tables I, III and IV for solar cells manufactured according to this specification shall be performed according to test methods and under test conditions stated below, unless otherwise specified in the applicable specification sheet.

4.5.2 Examination of product. The solar cells shall be examined to determine conformance to this specification with respect to materials, size, workmanship, or other requirements specified herein for which no tests are specified.

4.5.3 Material characterization. The following material parameters shall be determined by methods chosen by the supplier and in common usage by the industry on each full or partial ingot from which solar cells are to be manufactured.

a. Dopant material, level and original form

b. Crystal orientation

c. Resistivity

4.5.4 Contact integrity after high temperature. The solar cells shall be subjected to the following tests:

a. Perform electrical output test (4.5.9.2)

b. The solar cell shall be heated in an oven to a temperature specified in 3.4.5.5 and maintained at this temperature for at least 15 minutes. The solar cell shall then be allowed to naturally cool to room temperature.

c. Perform electrical output test (4.5.9.2)

d. Calculate percent degradation between a and c per 4.5.4.2.

4.5.4.1 Tape peel tests. Scotch Brand Magic Transparent Tape No 810 or equivalent adhesive tape with an adhesion-to-steel value of at least 336 grams per cm width in accordance with FED-STD-147 shall be placed over the cell to completely cover the cell surface. The tape shall be firmly rubbed until the cloudy appearance of the tape disappears, assuring firm adherence of the tape to the cell. The tape shall then be stripped from the cell at a 45 to 90 degree angle to the cell surface starting at the trailing edge of the gridline, peeling toward the contact pad edge of the cell. Following the peeling of the tape, the contact area shall be inspected for conformance with para 3.4.5.4.

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4.5.4.1.1 Front side. The solar cell without a cover slip shall be placed front side up on a flat surface for the tape peel test.

4.5.4.1.2 Back side. The solar cell shall be placed back side up on a flat surface forthe tape peel tests. On cells that have approximately the same contact void area on each long edge, the tape shall cover the entire cell surface.

4.5.4.2 Electrical output degradation. The current output I at constant voltage (at or near the maximum power point) shall be read to 0.1 mA/cm^2 precision. The electrical output data before and after the environment exposure shall be tabulated. This data shall contain the cell short circuit current (I_{sc}), the open circuit voltage (V_{oc}), and the current at the constant voltage (I_p). The average of the I_{sc} - after test and the I_{sc} - before test shall be determined. From these data the average ΔI shall be found. This ΔI shall be applied to all I_{sc} - after test and I_p - after test readings for light intensity normalization. The same process shall be repeated for the V_{oc} data. The ΔV reading shall be applied to the constant voltage - after test for cell temperature normalization. The normalized cell power values P (I_p x V_p) shall be determined individually and averaged for comparison to the requirements of 3.4.5.3.

4.5.5 Contact integrity after soldering or welding. The requirements of 3.4.5.6, 3.4.5.7, and 3.4.5.8 shall be verified by performing all of the following tests in the sequence given.

4.5.5.1 In-process contact test sequence. This paragraph is applicable only for solar cells selected according to sampling plan IP-2, table III and readied for the contact integrity test (test sequence 2 of table III.) Each cell shall be exposed to the following tests in the sequence given below. (If necessary, carefully bend pull tabs so that interference with the electrical test fixture and illuminating light source is minimized.)

a. Perform electrical output test (4.5.9.2)

b. Attach pull tabs to cells (4.5.5.4 or 4.5.5.5.)

c. Perform electrical output test (4.5.9.2)

d. Calculate percent degradation between a and c (4.5.4.2)

e. Perform nonlimited 45-degree pull test (4.5.5.6)

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4.5.5.2 Contact test sequence for qualification and quality conformance testing. This paragraph is applicable to solar cells selected for Group C qualification tests and sampling plan A quality conformance tests and readied for the contact test (test sequence 5 in table I and sequence 7 in table IV). Each cell shall be exposed to the tests in the sequence below. (If necessary, carefully bend pull tabs so that interference with the electrical test fixture and illuminating light source is minimized.)

a. Inspect contacts (4.5.5.3)

b. Measure electrical output (4.5.9.2)

c. Attach 4 tabs to cells (4.5.5.5)

d. Inspect contacts (4.5.5.3)

e. Repeat electrical output measurement (4.5.9.2)

f. Calculate degradation between b and e (4.5.4.2)

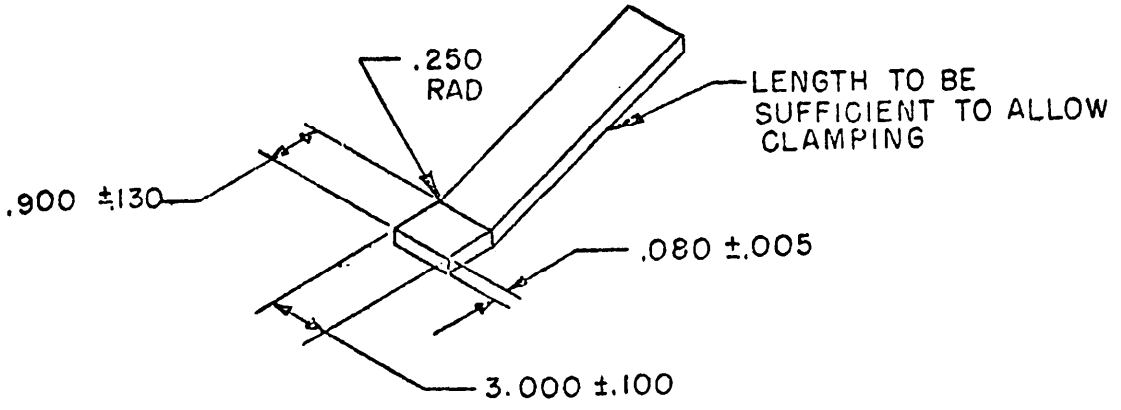
g. Perform nonlimited 45-degree pull test (4.5.5.6)

4.5.5.3 Contact appearance. Compliance with the requirements of 3.4.5.4 shall be verified by visual inspection using a stereo microscope of 5 to 10 power magnification or an episcopic comparator with equivalent magnification.

4.5.5.4 Pull tab soldering and test procedure. Solder four pull tabs to each solar cell to be tested, 2 on front contact areas and 2 on the back contact. Each pull tab shall be a No. 26 stranded pretinned wire bent approximately 90° for a length of 3 to 6 mm. The wire strands shall be clearly visible after soldering and MIL-S-45743 shall be used as a guide for soldering.

4.5.5.5 Welding of pull tabs. Four pull tabs shall be welded to each solar cell to be tested (2 on the front contact, 2 on the back.) Each tab shall be a flat ribbon of 3.0 $\pm$0.1 mm width, 75 $\pm$5µm thickness, and sufficient length for clamping in the pull tester as shown on figure 4. This welding shall be performed by the manufacturer with any welding method selected by the manufacturer and according to the guidelines given in 6.6.

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NOTES: 1. Tab foil material to be as specified in the applicable specification sheet

2. All dimensions in mm.

Figure 4. Contact Pull Tab

4.5.5.6 Nonlimited 45-degree pull test

4.5.5.6.1 Test procedure

a. Pull tabs shall be soldered or welded according to 4.5.5.4 or 4.5.5.5.

b. The cell shall be clamped in the pull tester described in 6.6.2.1. The tabs shall be bent up 45 degrees and clamped. The pull force shall be applied to the tabs at an angle of 45 degrees to the face of the cell (the angle enclosed by the two legs of the tab is 135 degrees) at a rate not exceeding 100 grams per second.

c. The reading of the pull force indicator shall be recorded at the instant of separation of the tab from the cell.

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d. For acceptability pull force shall exceed the applicable requirement of 3.4.5.6 (soldered) or 3.4.5.7 (welded).

4.5.6 Durability of anti-reflective surface coating. The solar cell without cover slip shall be placed in a beaker of boiling deionized water for 15 minutes. The solar cell shall then be removed and allowed to naturally cool to room temperature, It shall then be subjected to 20 complete cyles of eraser abrasion over the same area of anti-reflective coating with a continuous pressure of between 17 and 21.3 pounds per square inch (1.2 to 1.5 kg per square cm). During the test, the eraser shall be normal to the face of the solar cell. The eraser face shall be clean, and shall be A. W. Faber type No. 7107, alternatively Frankfort Arsenal eraser, or equivalent. The cell shall then be examined visually for defects in accordance with 3.4.2.1. Examination will be made under 10X magnification. Scuffing or scratching of the anti-reflective coating resulting from the test shall not constitute cause for rejection.

4.5.7 Visual inspection. Solar cells shall be inspected with the unaided eye to verify the respective requirements.

4.5.7.1 Verification of visual inspection. The solar cells shall be examined visually using a stereo microscope of 10X magnification or an optical comparator to verify the respective requirements.

4.5.8 Dimensions

4.5.8.1 Overall dimensions. The overall dimensions of the cells shall be verified by seating the cells flat within a cavity of dimensions given in the specified figure 1, 2 or 3. An optical comparator with overlay dimensions may be used to verify cover glass size and position.

4.5.8.2 Contact dimensions. The cell contact area dimensions shall be measured with an optical comparator, measuring microscope, or similar instrument.

4.5.8.3 Contact and gridline thickness and finish. Maximum contact thickness shall be measured perpendicular to and from the front surface of the cell with a microprofilometer or similar instrument.

4.5.9 Electrical output

4.5.9.1 Electrical current output test method. The electrical current output shall be measured with the light source of 4.5.9.3 at the constant voltage level, cell temperature, and light intensity specified in the applicable specification sheet. The cells shall be grouped into current output groups specified in 3.5.1.

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4.5.9.2 Electrical I-V curve output test method. The solar cell current-voltage characteristic shall be measured by illuminating the cell with the light source of 4.5.9.3 varying the load resistance across the cell, and plotting corresponding current-voltage data points. Short-circuit current is defined as current through a 1-ohm load, or lower. Open-circuit voltage is defined as voltage at a current of 1 milliampere, or less.

a. Unless otherwise directed, the following measurement data may be used in lieu of 4.5.9.2

(1) Short circuit current, I_{sc}

(2) Voltage at maximum power, V_{mp}

(3) Current at maximum power, I_{mp}

(4) Open circuit voltage, V_{oc}.

4.5.9.3 Light source. The light source shall conform to the requirements of 4.5.18. This conformance shall be demonstrated as follows:

4.5.9.3.1 Spectrum. The spectral content of the light shall be determined by at least one of the methods below:

a. Direct measurement with a calibrated spectro-radiometer

b. Indirect (comparison) measurement with a number of sensors of different spectral response characteristics.

4.5.9.3.2 Intensity. The light intensity at the test plane (plane of front side of solar cell) shall be determined by measuring the short-circuit currents of at least two standard solar cells as defined in 4.5.9.3.3. The intensity shall be adjusted so that the sum of the measured short-circuit currents of the standard cells used for calibration is equal to the sum of the calculated short-circuit currents for the light intesity specified in the applicable specification sheet. The deviation in measured short-circuit current for any one standard cell from the calculated value shall not exceed 1 percent.

4.5.9.3.3 Standard solar cells. All standard solar cells shall have the same characteristics as those solar cells which are to be delivered under this specification. The spectral similarity between the standard cells and production cells shall be demonstrated using xenon-tungsten ratios or similar generally employed methods. The standard cells will either be supplied by the procuring activity or mutually agreed upon by the manufacturer and the procuring activity.

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4.5.10 Magnetic field test. The solar cell shall be magnetized in a 2.5 millitesla magnetic field along its axis normal to the collector strip and in the plane of the cell. After this test, the magnetic field of the solar cell shall be measured in three mutually perpendicular planes at a distance from the cell specified in the applicable specification sheet.

4.5.11 Temperature cycling. The solar cells shall be placed in a chamber at ambient temperature and pressure. The chamber temperature shall then be varied to alternately bring the cells to the upper and lower temperature limits specified in 3.5.4. The dwell times at the temperature limits shall be not less than 2 minutes each. The time-rate of temperature change shall be within the limits specified in 3.5.4. The number of cycles is also specified in 3.5.4. The solar cell temperatures shall be monitored during this test with suitable thermoelectric instrumentation.

4.5.12 Storage temperature and humidity. The solar cells shall be placed in a chamber at ambient pressure. The chamber temperature and humidity shall then be brought to the levels specified in 3.5.3 and maintained for 30 days.

4.5.13 Particulate irradiation. The solar cell shall be subjected to the spectrum and fluences of particle radiation as specified in the applicable specification sheet. The flux density shall be uniform over the area of the solar cell to within $\pm$10 percent. During the radiation, the cell shall be kept at approximately 25°C, but not exceeding 29°C. Immediately following irradiation, the test specimen shall be stored at the temperature and for the recovery time period specified in the applicable specification sheet. After this time period, the solar cell shall be tested at 28°C in accordance with 4.5.9. The output levels observed after irradiation shall meet or exceed those indicated in the applicable specification sheet.

4.5.14 Weight. The average solar cell weight shall be determined by weighing samples of cells and dividing the group weight by the number of cells in the group. The weighing accuracy shall be $\pm$1.0 percent.

4.5.15 Electrical output before cover glass installation

a. After completion of cell fabrication through all steps in the PID prior to adhesive (if required) and cover glass application, the electrical output will be measured as required in 4.5.9.1.

b. Following this test cover glass will be applied to these same cells using the cover glass and adhesive and curing cycle (if required) in the applicable specification sheet. The electrical performance measurement will be repeated. The change in performance will be within the limits described in the specification sheet.

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4.5.16 Radiometric properties. Cover glasses shall be attached to solar cells as specified in 4.5.15b. After curing of adhesive and cleaning the radiometric properties of the covered front sides of cells shall be measured as follows:

4.5.16.1 Solar absorptance. The spectral reflectance of electromagnetic energy shall be measured from 280 to 2500 nanometers with Edwards type integrating sphere equipment. The test data shall be integrated over the solar irradiance and the integral subtracted from unity to yield the solar absorptance.

4.5.16.2 Hemispherical emittance. The hemispherical emittance shall be measured by either one of the two methods below:

a. Measure spectral reflectance in the wave-length range from 2.0 to 26 micrometers and integrate over 300° K Plankian radiator function.

b. Measure power density required to maintain sample at an equilibrium temperature of 300°K in a liquid nitrogen-cooled vacuum chamber.

4.5.17 High-temperature vacuum. The solar cells shall be placed in a vacuum chamber with pressure reduced to 1×10^{-5} Torr for a period of 168 hours and temperature of 140°C. At the end of this period the cells shall be allowed to return to room temperature and the electrical performance measured in accordance with 4.5.9.2.

4.5.18 Light source requirements for solar cell testing. Electrical output measurements shall be performed in natural or artifical sunlight which approximates natural sunlight under air-mass zero (AMO) conditions. The light source requirements shall be demonstrated by test in accordance with 4.5.9.3.

4.5.18.1 Spectral distribution. The AMO solar spectrum shall be defined by the figure entitled Solar Spectral Irradiance, ASTM E-490, in the range from 300 to 1200 nanometers. The total energy in any wavelength band incident on the test plane shall not deviate from the same band of the figure referenced above by more than the following percentage:

Wave Length Increment	±Percent
20 nanometers	30
40 nanometers	25
60 nanometers	20
100 nanometers	10

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4.5.18.2 Intensity uniformity. The intensity of any wavelength band specified in 4.5.18.1 shall not vary by more than ±2 percent across an area which is at least as large in any direction as the cell under test (or the standard cell, if larger), plus twice the sum of the cell positioning accuracy and the uniformity test cell diagonal dimension, when measured with a solar cell chip not larger than 3 x 5 mm in size.

4.5.18.3 Intensity stability. The intensity shall be stable within ±1 percent during any measurement period. If intensity-compensating measuring circuits are used, their speed of response shall be sufficiently fast to meet this stability requirement.

4.5.18.4 Intensity measurement. Intensity shall be measured by taking short-circuit current readings of silicon solar cells. Intensity calibrations shall be made with standard solar cells as defined in 4.5.9.3.3.

4.5.18.5 Ripple. The peak-to-peak ripple of the short-circuit current shall be less than 0.5 percent.

4.5.19 Inspection of preservation, packaging, and marking. Inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified herein. Containers with visual damage shall be rejected. The reading displayed by the visual humidity indicator card shall be recorded at the time of acceptance. In the event that the humidity indicator shows a humidity level greater than 35 percent, the units shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The solar cells shall be shipped in containers of a design approved by the procuring activity. The containers shall have adequate protective packing materials to prevent damage or shock of the solar cells. Each container shall be heat-sealed in an airtight, water-vapor-proof barrier bag of material in accordance with MIL-C-22191, together with sufficient bagged, activated, dry desiccant in accordance with MIL-D-3464 to prevent atmospheric or other contamination of the surface of the solar cells to assure compliance of cells to 3.3.5.

5.1.1 One unit of dry desiccants is required for each 90 square inches of bag area (one side only). This material shall not adversely affect the mechanical or electrical properties of the cells and shall not come into immediate contact with the cells.

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5.1.2 A card-type humidity indicator scaled to 10 to 80 percent relative humidity shall be placed within each sealed bag in an easily viewed location. This indicator shall be inspected in accordance with 4.5.19.

5.1.3 Any packing material used shall not affect the mechanical or electrical properties of the cells and shall not contaminate the surface of solar cells. The packaging method shall be approved by the procuring activity.

5.2 Marking of shipping containers. Each container shall be clearly identified by the following:

a. Manufacturer's name and specification number

b. Shipping lot number and quantity of cells

c. Month and year of manufacture

d. Cell current measured at a specified voltage. Current to be in current groups as given in the applicable specification sheet.

5.3 Deliverable documents. Each shipment of solar cells shall be accompanied by complete major inspection records of all examinations and test as specified in section 4 pertaining to the lot from which the cell shipment is made.

6. NOTES

6.1 Intended use. The solar cells with cover slips, covered by this specification are intended for use on solar panel assemblies for space flight applications. They are mounted on nonmetallic substrates or on insulated materials over a metallic substrate. Cells are electrically connected into solar energy converters for space applications. The adhesives and cover glass characteristics are specified in the applicable specification sheet.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, date of this specification and specification sheet

b. Type of solar cell, contact, and transparent cover

c. Level of preservation, packaging, and packing required

d. Samples subjected to destructive testing are not to be considered or shipped as part of the contract.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contacts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Air Force Aero Propulsion Laboratory, ATTN: POE-2, Wright-Patterson Air Force Base, Ohio 45433 and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions

6.4.1 Chips. Chips are defined as local absence of silicon along the edges and corners of a complete solar cell and which extend through the thickness of the cell.

6.4.2 Nicks. Nicks are defined as local absence of silicon along the edges and corners of a complete solar cell and which do not extend through the thickness of the cell.

6.4.3 Pinholes. Pinholes are defined as holes which penetrate surface layers (such as aluminum, humidity-protective or anti-reflective coating) to the silicon.

6.4.4 Voids. Voids are defined as the absence of deposited materials (such as aluminum or anti-reflective coating) of any dimension greater than 0.13 mm (0.005 inch).

6.4.5 Blisters. Blisters are defined as the partial or complete separation of materials at their interfaces as evidenced by local delamination.

6.4.6 Calculated current value. The calculated value of the short circuit current is that current expected in the specific setup being used to determine solar cell performance. This value may be based on balloon flight data or current measured under a different source at other than the test temperature.

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6.4.7 Electrical performance symbols. Electrical performance symbols and their meaning are as follows:

I_{sc} - Short circuit current
V_{oc} - Open current voltage
I_p - Current at constant voltage
ΔI - Change in ISC after test
ΔV - Change in VOC after test
V_{mp} - Voltage at maximum power
I_{mp} - Current at maximum power.

6.4.8 Fabrication lot. A fabrication lot is defined as that group of cells wherein conditions of contact applications were identical as nearly as practicable. Only cells with contacts applied within a 24 hour period may be grouped into a fabrication lot. Fabrication lots will be uniquely numbered.

6.4.9 Shipping lot. A shipping lot should consist of units manufactured under essentially the same conditions and at the same time. The definition of a lot is in accordance with paragraph entitled Inspection lot of MIL-S-19500. All shipping lots shall be consecutively numbered. A shipping lot should contain all members of a fabrication lot.

6.4.10 Destructive tests. The following tests are considered destructive:

Contact integrity, temperature (4.5.4)

Contact integrity, welding (4.5.5)

Anti-reflective surface coating (4.5.6)

Humidity and temperature (4.5.12)

Particle irradiation (4.5.13)

Electrical output before cover glass installation (4.5.15)

6.4.11 Synthetic oils. Any synthetic oil, grease, or substance including silicone oils and grease, which may prevent the use of adhesives specified in the applicable specification sheet, should not come in contact with any portion of the solar cell. Such oils and grease cannot adequately be cleaned from solar cells once applied. Natural oils are permissible.

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6.5 Cover glass installation. Cover glass installation should be performed as described in the applicable specification sheet.

6.6 Welding procedure for attaching pull tabs to cells

6.6.1 Welding equipment. As a guide, the ultrasonic bonder electrode may be tungsten carbide with a stylus 1.6 mm in diameter having a 5.0 mm approximately spherical tip radius.

6.6.2 Welding schedule. The welding shedule will be at the manufacturer's option.

6.6.2.1 Pull tester. The following equipment manufactured by Unitek Corporation, Monrovia, California, may be used for the pull test of 4.5. 6.

a. Pull test, model 6-092-01:

b. Force gage (25 newtons, 5 pounds) model 7-015-01 (Chatillon gage).

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